Project: Worked examples
Title: Column base plate (EN 1993-1-8)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

An unstiffened rectangular base plate under an I/H, RHS or CHS column on grouted concrete: bearing resistance, the force regime and holding-down bolt tension for a major-axis moment, the tension T-stub of the plate and the compression T-stub, shear by friction and anchor bolts, and the plate thickness that satisfies them together. The bolt tension and shear per anchor go to Hilti PROFIS for the anchorage, which is done outside this sheet.
Bearing (cl. 6.2.5): f_cd = α_cc·f_ck/γ_c; f_jd = β_j·α·f_cd, β_j = 2/3; c = t_p·√[f_y/(3·f_jd·γ_M0)] Axial (cl. 6.2.8.2): N_j,Rd = f_jd·A_eff, A_eff the bearing strips of width c round the column footprint (full rectangle when they overlap); u_N = N_c,Ed/N_j,Rd Moment (cl. 6.2.8.3, Table 6.7): z_C from the section ((h − t_f)/2, (h − t)/2, half-annulus centroid) or entered, x_b the bolt row from the axis, z = z_C + x_b; C_Ed = (M_Ed + N_Ed·x_b)/z and T_Ed = C_Ed − N_Ed when both are ≥ 0; with N_Ed > 0 and T_Ed < 0 both sides are in compression, C = N_Ed/2 ± M_Ed/(2z_C); with N_Ed < 0 and C_Ed < 0 both rows are in tension, T = −N_Ed/2 ± M_Ed/(2x_b); F_t,Ed = T_Ed/n_t at the more loaded row Plate T-stub (cl. 6.2.6.11, 6.2.4, Table 6.2): M_pl,Rd = 0.25·l_eff·t_p²·f_y/γ_M0; without prying min(2·M_pl,Rd/m, n_t·F_t,Rd); with prying the modes 4·M_pl,Rd/m, (2·M_pl,Rd + n·ΣF_t,Rd)/(m + n) with n = min(e_min, 1.25m), and n_t·F_t,Rd; l_eff is the total for the n_t bolts Compression T-stub (cl. 6.2.6.9, 6.2.6.10): C_Rd = f_jd·A_flange, checked when M_Ed > 0 and the resultant is in compression Shear (cl. 6.2.2): F_2,vb,Rd = α_bc·f_ub·A_s/γ_M2, α_bc = 0.44 − 0.0003·f_yb; F_vb,Rd = min(F_1,vb,Rd, F_2,vb,Rd); V_Rd = C_f,d·N_c,Ed + n_b·F_vb,Rd; V_b,Ed = max(V_Ed − C_f,d·N_c,Ed, 0)/n_b Required thickness: the smallest t_p in 10–100 mm with u_N, u_C and u_T all ≤ 1 (each falls as t_p rises)

Ref: EN 1993-1-8:2005 cl. 6.2.2 — Design resistance of column bases in shear
Ref: EN 1993-1-8:2005 cl. 6.2.4, Table 6.2 — Equivalent T-stub in tension
Ref: EN 1993-1-8:2005 cl. 6.2.5 — Equivalent T-stub in compression
Ref: EN 1993-1-8:2005 cl. 6.2.6.9 — Concrete in compression including grout
Ref: EN 1993-1-8:2005 cl. 6.2.6.10 — Base plate in bending under compression
Ref: EN 1993-1-8:2005 cl. 6.2.6.11 — Base plate in bending under tension
Ref: EN 1993-1-8:2005 cl. 6.2.6.12 — Anchor bolts in tension
Ref: EN 1993-1-8:2005 cl. 6.2.8.2 — Column bases subject to axial forces
Ref: EN 1993-1-8:2005 cl. 6.2.8.3, Table 6.7 — Column bases subject to axial force and bending moment
Ref: EN 1992-1-1:2004 cl. 3.1.6, 6.7 — Design strength of concrete; partial area loading

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| column | Column section type | I/H | | drawing: PF-GA-001 rev C | any |
| h | Column depth, along the plate length L (I/H and RHS) | 260.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 50 … 1200 mm |
| b | Column width, along the plate width B (I/H and RHS) | 256.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 50 … 1200 mm |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| t_f | Flange thickness (I/H) | 17.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 3 … 100 mm |
| t_w | Web thickness (I/H) | 10.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 3 … 100 mm |
| B | Plate width | 400 | mm | drawing: PF-BP-001 rev B | 100 … 1500 mm |
| L | Plate length | 450 | mm | drawing: PF-BP-001 rev B | 100 … 1500 mm |
| t_p | Plate thickness | 25 | mm | drawing: PF-BP-001 rev B | 10 … 100 mm |
| f_y | Plate yield strength | 265 | MPa | spec: BS EN 10025-2 S275, 16 mm < t ≤ 40 mm | 235 … 460 MPa |
| f_ck | Concrete characteristic cylinder strength | 30 | MPa | spec: BS EN 206 C30/37 pad foundation, drawing PF-FN-001 | 20 … 50 MPa |
| f_ck_g | Grout characteristic strength (optional: checks the β_j = 2/3 condition) | 60 | MPa | spec: Non-shrink grout datasheet, 28-day strength (placeholder GR-DS-01) | 5 … 100 MPa |
| t_g | Grout thickness (optional: checks the β_j = 2/3 condition) | 30 | mm | drawing: PF-BP-001 rev B | 1 … 200 mm |
| alpha_cc | α_cc, long-term effects on the concrete strength | 0.85 | | spec: UK NA to BS EN 1992-1-1, NA to cl. 3.1.6(1)P | 0.8 … 1 |
| gamma_c | γ_c, partial factor for concrete | 1.5 | | spec: UK NA to BS EN 1992-1-1, Table NA.1 | 1.2 … 1.8 |
| gamma_M0 | γ_M0, partial factor for the plate | 1 | | spec: UK NA to BS EN 1993-1-1, NA.2.15 | 1 … 1.25 |
| gamma_M2 | γ_M2, partial factor for the anchor bolts | 1.25 | | spec: UK NA to BS EN 1993-1-8, NA.2.3 | 1.1 … 1.5 |
| C_fd | C_f,d, friction coefficient plate to grout | 0.2 | | spec: EN 1993-1-8 cl. 6.2.2, sand-cement grout | 0 … 0.45 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| alpha | α, concentration factor √(A_c1/A_c0) of the foundation | 1.5 | | assumption: Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001. | 1 … 3 |
| alpha_basis | Foundation geometry that α rests on (state it when α > 1) | Pad 1.2 × 1.2 m, 0.8 m deep (PF-FN-001): A_c1/A_c0 above 2.25, so α = 1.5 is taken | | drawing: PF-FN-001 rev A | any |
| N_Ed | Axial force, compression positive | 210 | kN | reference: S5 load combinations sheet, ULS 6.10b, wind leading, column base A | -5000 … 10000 kN |
| M_Ed | Major-axis moment (uniaxial) | 95 | kN*m | reference: S5 load combinations sheet / S2 frame sheet, column base A | 0 … 2000 kN*m |
| V_Ed | Shear force | 60 | kN | reference: S5 load combinations sheet / S2 frame sheet, column base A | 0 … 2000 kN |
| n_t | Holding-down bolts in each bolt row (one row each side) | 2 | | drawing: PF-BP-001 rev B | 1 … 20 |
| n_b | Holding-down bolts in total (both sides) | 4 | | drawing: PF-BP-001 rev B | 2 … 40 |
| x_b | Bolt row distance from the column axis (outside the column face, inside the plate edge) | 185 | mm | drawing: PF-BP-001 rev B | 50 … 1000 mm |
| m | Bolt to flange-weld toe distance, tension T-stub | 46.9 | mm | assumption: Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm. | 5 … 300 mm |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| l_eff | Effective length of the tension T-stub, total for the n_t bolts (cl. 6.2.6.5 patterns) | 350 | mm | assumption: Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate). | 10 … 2000 mm |
| prying | Prying forces develop (short bolts, L_b ≤ L_b*); off is conservative for the plate | no | | assumption: Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case). | any |
| F_t_Rd | Tension resistance per bolt (from S4) | 203 | kN | reference: S4 bolt sheet, M24 8.8 (203.04 kN), rounded down | 1 … 2000 kN |
| f_ub | Anchor bolt ultimate strength | 800 | MPa | spec: EN ISO 898-1 class 8.8 | 400 … 1000 MPa |
| f_yb | Anchor bolt yield strength (α_bc is valid for 235–640 MPa) | 640 | MPa | spec: EN ISO 898-1 class 8.8 | 235 … 640 MPa |
| A_s | Anchor bolt tensile stress area (from S4) | 352.5 | mm^2 | reference: S4 bolt sheet, M24 stress area (352.50 mm²) | 50 … 1500 mm^2 |
| F_1_vb_Rd | Bolt bearing resistance on the plate, F_1,vb,Rd (from S4 F_b,Rd) | 364.4 | kN | reference: S4 bolt sheet, M24 bearing on 25 mm S275 plate, $e_1$ = 60 mm, $d_o$ = 27 mm (364.44 kN) | 1 … 2000 kN |
| profis_ref | Hilti PROFIS report reference (the report is filed as Evidence) | EVD-PROFIS-PLACEHOLDER | | evidence: EVD-PROFIS-PLACEHOLDER | any |

### 3. Assumptions

A1. Static loading; a major-axis moment only (uniaxial); an unstiffened rectangular plate on grout of adequate strength and thickness for β_j = 2/3 (checked when the grout strength and thickness are entered). (Scope of S6.)
A2. The simplified T-stub model: one compression resultant at the centre of the compression T-stub (flange centre, or the half-annulus centroid of a CHS) and one tension resultant per bolt row; the force regime follows EN 1993-1-8 Table 6.7. The web is not given credit in the moment check. (EN 1993-1-8 cl. 6.2.8.3.)
A3. The effective length l_eff is the total for the n_t bolts of a row, read from the cl. 6.2.6.5 patterns; it is used for modes 1 and 2 alike (l_eff,1 ≤ l_eff,2, so this is conservative for mode 2). (One input.)
A4. Concrete, anchor tension, pull-out, breakout, edge and the tension–shear interaction of the anchors are designed in Hilti PROFIS from F_t,Ed and V_b,Ed per anchor; the report is filed as Evidence and referenced. With prying switched on the prying force Q is not in F_t,Ed and PROFIS must add it. (External anchorage design.)

A5. The friction resistance counts only the compression, C_f,d·max(N_Ed, 0); a column in net tension has none. Anchor bolts share shear equally, with their holes assumed no larger than the standard clearance (oversized or slotted holes are outside this sheet). (EN 1993-1-8 cl. 6.2.2.)
A6. The bearing outstand c is cut back where it would pass the plate edge; the web and the column interior keep the full c. The CHS compression side is half of the annular bearing area, with its centroid as lever arm (an approximation). (Conservative against an uncut c.)
A7. The compression T-stub is checked against the flange area alone as soon as M_Ed > 0, while at M_Ed = 0 the whole footprint (web included) resists, so the compression check jumps at M_Ed = 0⁺ (conservative). (Limitation.)
A8. Partial factors and the friction coefficient come from the records in use (γ_c, α_cc: EN 1992-1-1 NA; γ_M2: EN 1993-1-8 NA; γ_M0: EN 1993-1-1 NA; C_f,d: the grout used); the values shown default to the EN recommended and UK values. (§A5 small factor set.)
A9. alpha (α, concentration factor √(A_c1/A_c0) of the foundation) is taken as 1.5. (Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001.)
A10. m (Bolt to flange-weld toe distance, tension T-stub) is taken as 46.9 mm. (Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.)
A11. l_eff (Effective length of the tension T-stub, total for the n_t bolts (cl. 6.2.6.5 patterns)) is taken as 350 mm. (Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).)
A12. prying (Prying forces develop (short bolts, L_b ≤ L_b*); off is conservative for the plate) is taken as no. (Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case).)

## 4. Working

[31] f_cd = alpha_cc * f_ck / gamma_c — EN 1992-1-1 cl. 3.1.6
= 0.85 × 30 MPa / 1.5
= 17 MPa
Concrete design compressive strength

[32] beta_j = 2/3, fixed — EN 1993-1-8 cl. 6.2.5
= grout conditions of cl. 6.2.5 assumed met
= 0.66667
Joint coefficient β_j

[33] f_jd = beta_j * alpha * f_cd — EN 1993-1-8 cl. 6.2.5
= 0.66667 × 1.5 × 17 MPa
= 17 MPa
Design bearing strength of the joint

[34] c = t_p * sqrt(f_y / (3 * f_jd * gamma_M0)) — EN 1993-1-8 cl. 6.2.5
= 25 mm × sqrt(265 MPa / (3 × 17 MPa × 1))
= 56.987 mm
Additional bearing width round the column footprint

[35] A_eff = 2(b + 2c)(t_f + 2c) + (h - 2t_f - 2c)(t_w + 2c) — EN 1993-1-8 cl. 6.2.5
= 2 × 370.27 × 131.27 + 111.73 × 124.27 mm²
= 111100 mm²
Effective bearing area

[38] N_j_Rd = f_jd * A_eff — EN 1993-1-8 cl. 6.2.8.2, 6.2.5
= 17 MPa × 111100 mm²
= 1888.7 kN
Axial bearing resistance of the joint

[39] N_c_Ed = max(N_Ed, 0)
= N_Ed = 210 kN
= 210 kN
Design compression in the column

[40] u_N = N_c_Ed / N_j_Rd
= 210 kN / 1888.7 kN
= 0.11119
Axial bearing utilisation

[41] t_req_N = smallest t_p in 10–100 mm with f_jd·A_eff(c(t_p)) ≥ N_c,Ed (bisection) EN 1993-1-8 cl. 6.2.5
= N_c,Ed = 210 kN; carried at the 10 mm minimum
= 10 mm
Plate thickness required for axial bearing alone

[42] z_C = (h − t_f)/2 EN 1993-1-8 cl. 6.2.8.3
= (260.3 − 17.3)/2 mm
= 121.5 mm
Compression resultant distance from the column axis

[43] z = z_C + x_b EN 1993-1-8 cl. 6.2.8.3
= 121.5 + 185
= 306.5 mm
Lever arm between the compression resultant and the bolt row

[44] C_Ed = (M_Ed + N_Ed·x_b)/z EN 1993-1-8 cl. 6.2.8.3, Table 6.7
= (95 kN·m + 210 kN × 185 mm)/306.5 mm
= 436.7 kN
Compression resultant under the compression flange

[45] T_Ed = C_Ed − N_Ed EN 1993-1-8 cl. 6.2.8.3, Table 6.7
= 436.7 − 210
= 226.7 kN
Tension resultant at the bolt row

[46] C_Ed_2 = none
= other side
= 0 kN
Compression resultant under the other flange

[47] T_Ed_2 = none
= other row
= 0 kN
Tension resultant at the other bolt row

[48] F_t_Ed = T_Ed / n_t EN 1993-1-8 cl. 6.2.6.12
= 226.7 kN / 2
= 113.35 kN
Holding-down bolt tension per bolt at the more loaded row (→ PROFIS)

[49] A_flange = (b + 2c)(t_f + 2c), at most A_eff/2 EN 1993-1-8 cl. 6.2.6.9
= min(370.27 × 131.27, A_eff/2) mm²
= 48608 mm²
Bearing area of the compression flange

[50] C_Rd = f_jd * A_flange EN 1993-1-8 cl. 6.2.6.9, 6.2.6.10
= 17 MPa × 48608 mm²
= 826.33 kN
Bearing resistance of the compression flange

[51] u_C = C_Ed / C_Rd
= 436.7 kN / 826.33 kN
= 0.52849
Compression flange utilisation

[52] M_pl_Rd = 0.25 * l_eff * t_p^2 * f_y / gamma_M0 EN 1993-1-8 Table 6.2
= 0.25 × 350 mm × (25 mm)^2 × 265 MPa / 1
= 14.492 kN·m
Plastic moment resistance of the plate over l_eff

[53] T_Rd_1 = 2 * M_pl_Rd / m — EN 1993-1-8 Table 6.2
= 2 × 14.492 kN·m / 46.9 mm
= 618 kN
Mode 1-2 without prying forces

[54] T_Rd_3 = n_t * F_t_Rd — EN 1993-1-8 Table 6.2
= 2 × 203 kN
= 406 kN
Mode 3: bolt failure

[55] T_Rd = min(T_Rd_1, T_Rd_3) — EN 1993-1-8 cl. 6.2.4
= min(618 kN, 406 kN)
= 406 kN
Tension T-stub resistance (no prying)

[56] u_T = T_Ed / T_Rd
= 226.7 kN / 406 kN
= 0.55839
Plate bending and bolts in tension, T-stub utilisation

[57] u_bt = F_t_Ed / F_t_Rd
= 113.35 kN / 203 kN
= 0.55839
Holding-down bolt tension utilisation

[58] t_req = smallest t_p in 10–100 mm with u_N, u_C, u_T ≤ 1 (bisection) — EN 1993-1-8 cl. 6.2.5, 6.2.6.9, 6.2.6.11
= N_c,Ed = 210 kN, C_Ed = 436.7 kN, T_Ed = 226.7 kN
= 15.142 mm
Plate thickness required

[59] alpha_bc = 0.44 - 0.0003·f_yb — EN 1993-1-8 cl. 6.2.2
= 0.44 - 0.0003 × 640
= 0.248
Anchor bolt shear factor α_bc

[60] F_2_vb_Rd = alpha_bc * f_ub * A_s / gamma_M2 — EN 1993-1-8 cl. 6.2.2
= 0.248 × 800 MPa × 352.5 mm² / 1.25
= 55.949 kN
Anchor bolt shear resistance F_2,vb,Rd

[61] F_vb_Rd = min(F_1_vb_Rd, F_2_vb_Rd) — EN 1993-1-8 cl. 6.2.2
= min(364.4 kN, 55.949 kN)
= 55.949 kN
Anchor bolt shear resistance per bolt (bearing or shear)

[62] F_f_Rd = C_fd * N_c_Ed — EN 1993-1-8 cl. 6.2.2
= 0.2 × 210 kN
= 42 kN
Friction resistance, compression only

[63] V_Rd = F_f_Rd + n_b * F_vb_Rd — EN 1993-1-8 cl. 6.2.2
= 42 kN + 4 × 55.949 kN
= 265.8 kN
Shear resistance of the base

[64] u_V = V_Ed / V_Rd
= 60 kN / 265.8 kN
= 0.22574
Shear utilisation

[65] V_b_Ed = max(V_Ed - F_f,Rd, 0)/n_b — EN 1993-1-8 cl. 6.2.2
= max(60 - 42, 0)/4
= 4.5 kN
Shear per anchor bolt after friction (→ PROFIS)

[66] u_max = max(u_N, u_C, u_T, u_bt, u_V)
= u_N = 0.11119, u_C = 0.52849, u_T = 0.55839, u_bt = 0.55839, u_V = 0.22574
= 0.55839
Governing utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| f_cd | 17 MPa | Concrete design strength |
| beta_j | 0.66667 | Joint coefficient β_j |
| f_jd | 17 MPa | Bearing strength f_jd |
| c | 56.987 mm | Bearing outstand c |
| A_eff | 111100 mm² | Effective bearing area |
| N_j_Rd | 1888.7 kN | Axial bearing resistance |
| u_N | 0.11119 | Axial bearing utilisation |
| t_req_N | 10 mm | Plate thickness required for axial bearing alone |
| t_req | 15.142 mm | Plate thickness required (bearing, compression and tension T-stubs) |
| z_C | 121.5 mm | Compression lever arm z_C |
| z | 306.5 mm | Lever arm z |
| C_Ed | 436.7 kN | Compression resultant |
| C_Ed_2 | 0 kN | Compression resultant, other side |
| T_Ed | 226.7 kN | Tension resultant, more loaded row |
| T_Ed_2 | 0 kN | Tension resultant, other row |
| F_t_Ed | 113.35 kN | Bolt tension per bolt (to PROFIS) |
| A_flange | 48608 mm² | Compression-side bearing area |
| C_Rd | 826.33 kN | Compression-side resistance |
| u_C | 0.52849 | Compression-side utilisation |
| M_pl_Rd | 14.492 kN·m | Plate plastic moment |
| T_Rd_1 | 618 kN | T-stub mode 1-2, no prying |
| T_Rd_3 | 406 kN | T-stub mode 3 |
| T_Rd | 406 kN | Tension T-stub resistance |
| u_T | 0.55839 | Tension T-stub utilisation |
| u_bt | 0.55839 | Bolt tension utilisation |
| alpha_bc | 0.248 | Anchor shear factor α_bc |
| F_2_vb_Rd | 55.949 kN | Anchor shear resistance F_2,vb,Rd |
| F_vb_Rd | 55.949 kN | Anchor shear resistance per bolt |
| F_f_Rd | 42 kN | Friction resistance |
| V_Rd | 265.8 kN | Shear resistance |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| u_V | 0.22574 | Shear utilisation |
| V_b_Ed | 4.5 kN | Shear per anchor bolt (to PROFIS) |
| u_max | 0.55839 | Governing utilisation |
| regime | "tension at one bolt row, compression under the other flange (Table 6.7, z = z_C + x_b)" | Force regime (Table 6.7) |
| T_mode | "mode 3 (bolt failure)" | Governing T-stub mode |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Base plate tension T-stub (bending at the bolts) | u_T = 0.55839 | u_T ≤ 1 | 0.558 | EN 1993-1-8 cl. 6.2.6.11 |
| PASS | Holding-down bolt tension | u_bt = 0.55839 | u_bt ≤ 1 | 0.558 | EN 1993-1-8 cl. 6.2.6.12 |
| PASS | Compression flange bearing | u_C = 0.52849 | u_C ≤ 1 | 0.528 | EN 1993-1-8 cl. 6.2.6.9, 6.2.6.10 |
| PASS | Shear: friction plus anchor bolts | u_V = 0.22574 | u_V ≤ 1 | 0.226 | EN 1993-1-8 cl. 6.2.2 |
| PASS | Concrete bearing under the axial force | u_N = 0.11119 | u_N ≤ 1 | 0.111 | EN 1993-1-8 cl. 6.2.8.2, 6.2.5 |
| PASS | Anchorage designed externally (PROFIS report filed as Evidence) | PROFIS-PLACEHOLDER | PROFIS report bolt tension 113.35 kN and shear 4.5 kN per anchor designed in PROFIS | 0 | |

## 7. Applicability

WARNING — record.alpha_cc: alpha_cc: a tabulated value not taken from a reference record (EN 1992-1-1 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.gamma_c: gamma_c: a tabulated value not taken from a reference record (EN 1992-1-1 National Annex factors); enter it as one so its source, edition and licence travel with it.
WARNING — record.gamma_M2: gamma_M2: a tabulated value not taken from a reference record (EN 1993-1-8 National Annex factor); enter it as one so its source, edition and licence travel with it.
WARNING — record.gamma_M0: gamma_M0: a tabulated value not taken from a reference record (EN 1993-1-1 National Annex factor); enter it as one so its source, edition and licence travel with it.
WARNING — record.C_fd: C_fd: a tabulated value not taken from a reference record (Friction coefficient plate to grout); enter it as one so its source, edition and licence travel with it.
WARNING — record.f_y: f_y: a tabulated value not taken from a reference record (Plate strength); enter it as one so its source, edition and licence travel with it.

## 8. Verification

Template struct.base-plate@1.0.0 Content hash 5e6f7b62171087af Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0